



RECYCLING AND REUSING THE PLASTIC WASTE PRODUCED DURING 3-D PRINTING PROCESS

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RECYCLING AND REUSING
THE PLASTIC WASTE
PRODUCED DURING 3-D
PRINTING PROCESS

MOTIVATION OF THE PROJECT

- The motivation of our project is to evoke an eco-conscious approach during the process of 3D printing.
- We make sure to appeal that the filament extruders are very **useful machines** to promote sustainability by enabling the **recycling** of plastic **waste and failed** 3D prints.
- By reusing these materials, we reduce plastic waste and help make 3D printing more environmentally friendly.

Literature Survey

- We approached the **seniors** who had earlier worked on the project, assistant professors, research scholars, to gather information.
- We researched and read various **websites** that dealt with various aspects related to the project, like properties of PLA plastic, and various ways to cool the filament after it is made, etc.
- We also referred to the methods chosen by few successful makers of the project who have presented their ideas and workings through **YouTube**.

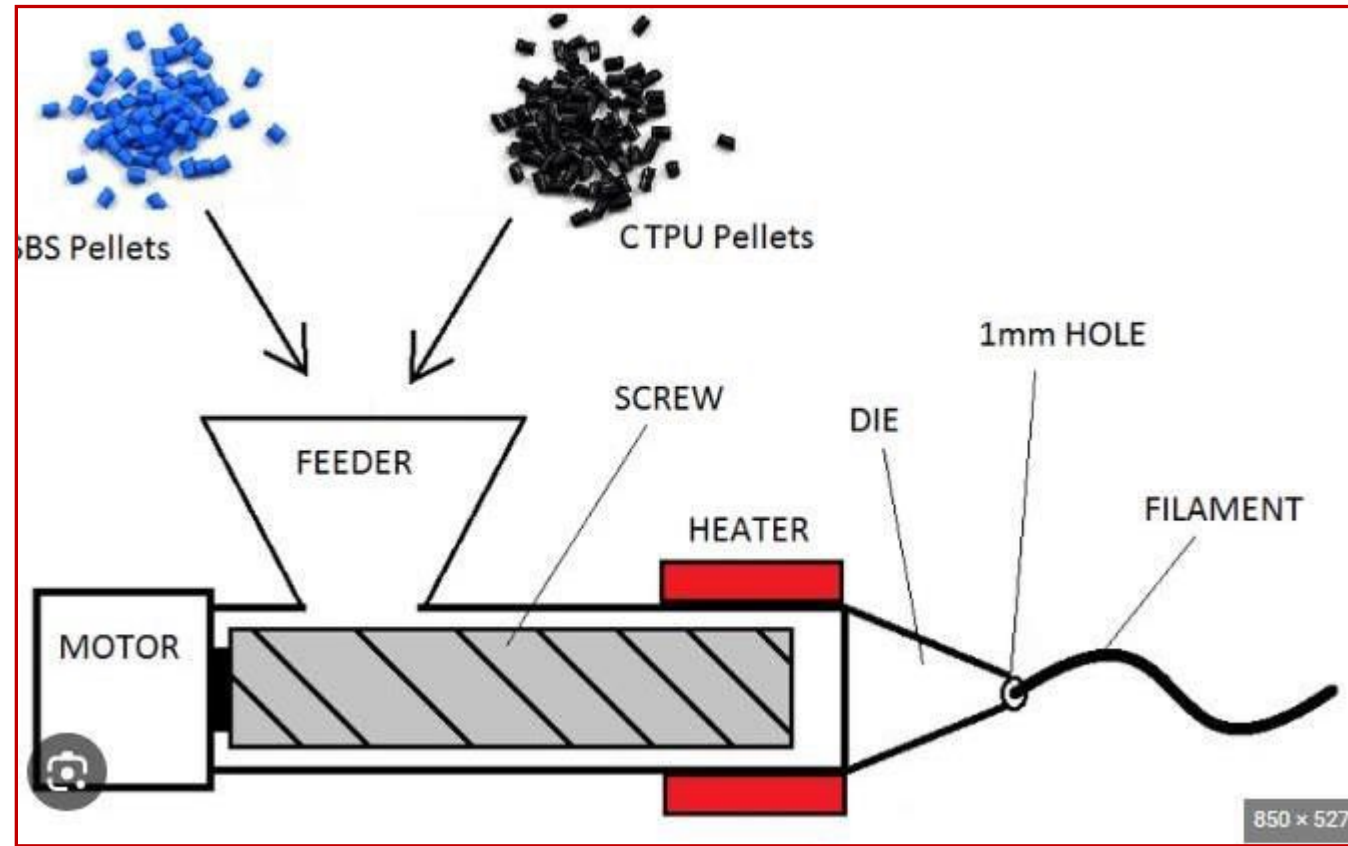
AIM:

TO RECYCLE THE PLASTIC SUPPORT STRUCTURE USED IN FDM (fused deposition modelling) 3-D PRINTERS, BY REMOLDING THEM AGAIN INTO FILAMENTS, WHICH CAN BE REUSED IN PRINTING, THUS REDUCING PLASTIC POLLUTION.

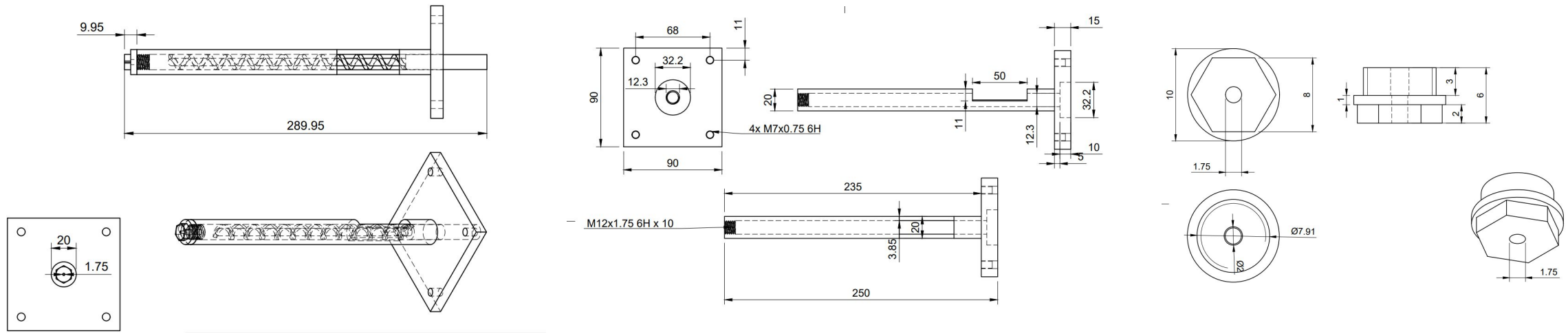
METHODOLOGY:

1. We Collect plastic waste (the support system) used for 3-D printing of a model.
2. Then we sort the waste and **clean it**.
3. Then, during the plastic pellets processing, the waste is **grinded** into small plastic pellets. (Here the pellets used for recycled filament is composed of PLA, i.e. polylactic acid)
4. During the plastic pellets processing stage, the plastic bits undergo **a melting process** in the extruder to form a molten mass.
5. Once the plastic pellets are in a molten state, they are **fed into the extrusion** barrel.
6. Next , in the filament formation process,there is a **die head which imparts shape** and diameter to the filament as it emerges from the extruder. The filament is cooled and **wound to on a spool** .
7. This filament can be used to print successive 3D models.

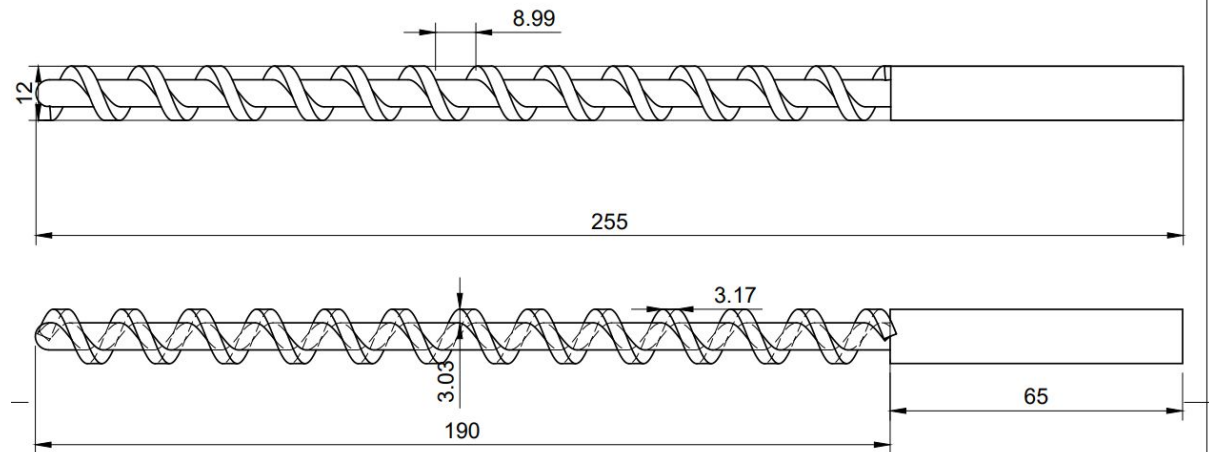
BLOCK DIAGRAM



Design of few components



245mm BARREL AND SCREW AND
NOZZLE
255mm SCREW
245mm BARREL
NOZZLE



List of components and their ratings

S.no	Required materials	Number of sets	cost	Total cost
1	Screw , Barrel and Nozzle .	1		
2	12V DC Square Gear/Geared Motor 200 RPM - High Torque	1	2000	2000
3	Display LED Motor Speed Controller	1	500	500
4	Band Nozzle Heater	2	1070	2140
5	Temperature Controller and Thermocouple Sensor	2	500	1000
6	AUTOMATION Solid State Relay SSR DC TO AC	2	380	760
7	Shaft Coupler	1	500	500

8	12V Cooling Fan	2	125	250
9	12V Dc 10 RPM Geared Motor	2	225	450
10	Motor Speed Regulator Control	2	210	420
11	12V 10A 120W DC Switching Switch Power Supply	1	800	800
12	Metal Shield screw Bearings (2) (ID=12mm, OD=32mm, Width=10mm)	1	250	250
13	Switches (5)	5	30	150
14	Click UP Rubber Roller	1	450	450
15	Small Bearings(8x22x7mm) (set 10)	1	300	300

16	12V motor Stand (2)	1	370	370
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List of components and their ratings

- ZIP lock bag 112 Rs
- PLA Filament 1kg roll white color 950 Rs
- 1Pcs 8x12mm Motor Jaw Shaft Coupler 500
- Wirings
- Nut and bolts (x4)
- Screws (x5) 50 Rs
- Plug



Status and progress of
project and the prototype
developed

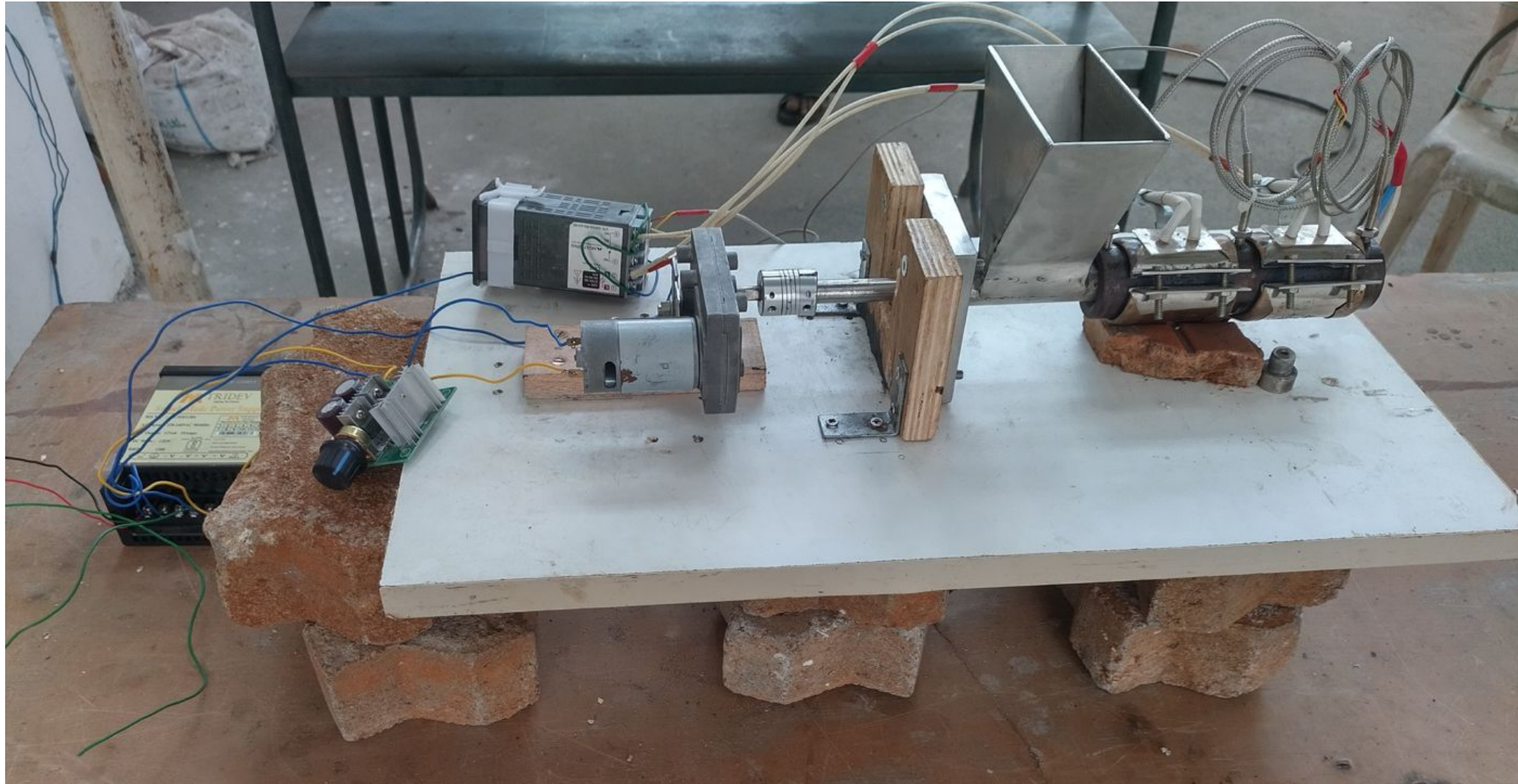
BACKGROUND

- Based on inputs from our seniors and our own analysis we got to know that the project was **partially completed** to an extent.
- Our seniors were successful in connecting the screw, feeder, heater, thermocouples to the temperature controller, AC to DC converter unit.
- The base of the motor was **made in 3-D printing**.

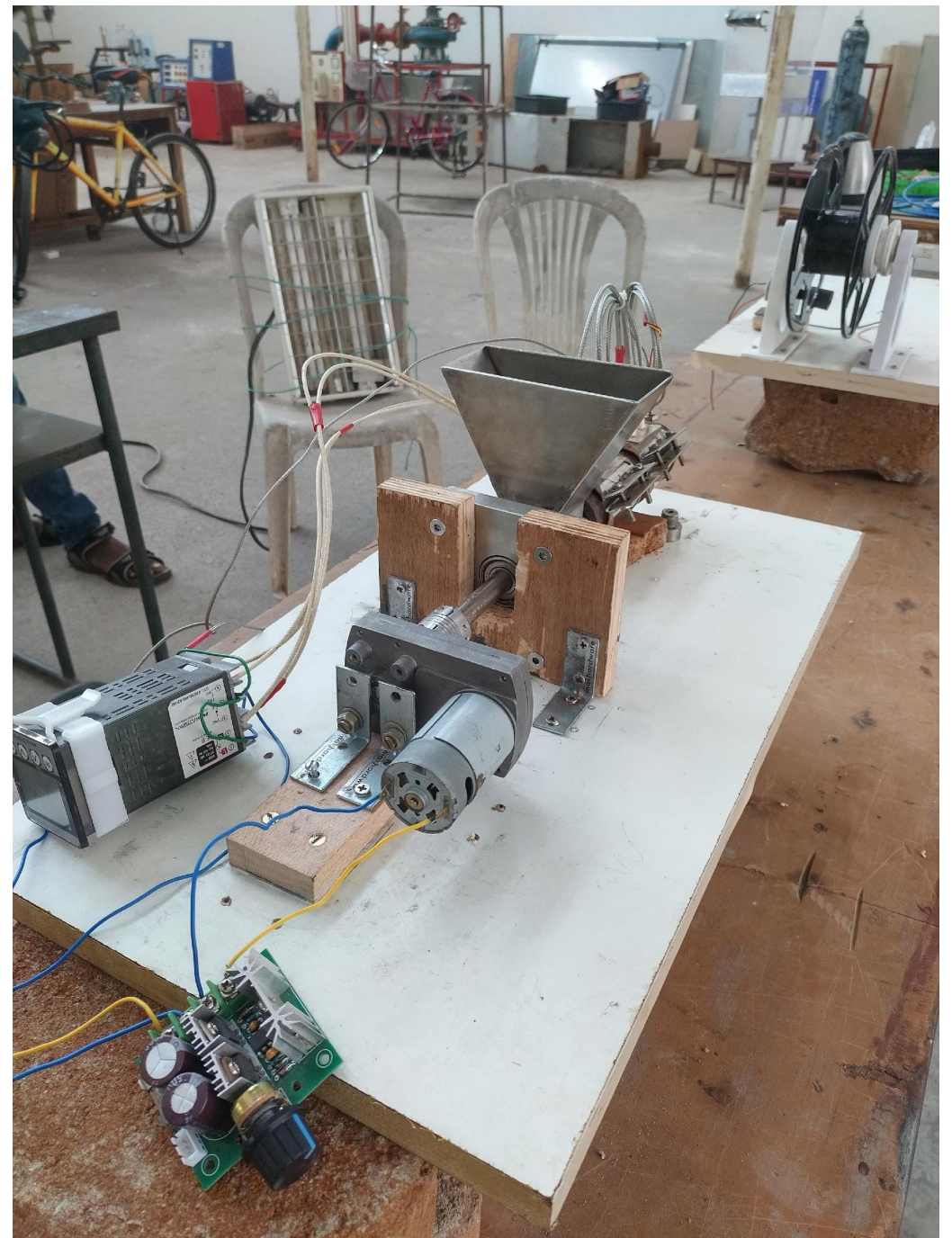
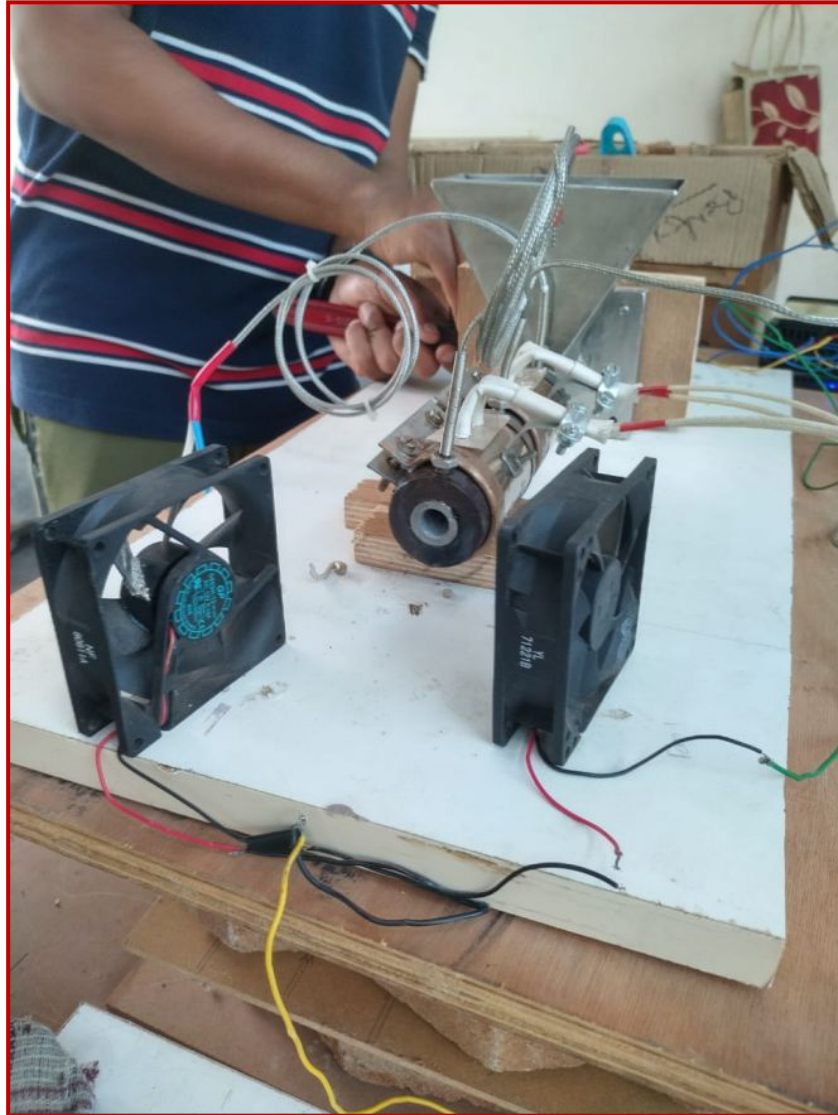
BACKGROUND

- The shaft did **not properly align** with the screw of the extruder because the thickness of the 3D printer base frame was **less than the optimum thickness** also since it was made of plastic it **did not possess strength** and durability.
- In addition the wooden block attached to the extruder block was not in proper position due to which **severe vibration** occurred which was causing **damage to motor and coupler**.
- Due to this misalignment, the **coupling failed twice** which added expenses to our project.
- Due to the misalignment of the wooden block, **fitting of screws could not be done properly** with the metal support plate, due to which the **machine was loose** and not sturdy.
- Due to the above misalignment problems, plastic pieces charged inside the funnel got stuck.
- Due to this the **shaft got seized**.

Assembly of the set-up



Assembly of the set-up



OUR WORK AS A GROUP

- As a group, we analysed the project.
- We distributed the work among ourselves.
- Some of us we did literature survey, approached the seniors, professors, research scholars, to gather further information.
- Many hardware items were to be bought to start the activities.
- We took around three to four days to gather all the items.
- When we started it looked too complicated and we faced many problems.

OUR WORK AS A GROUP

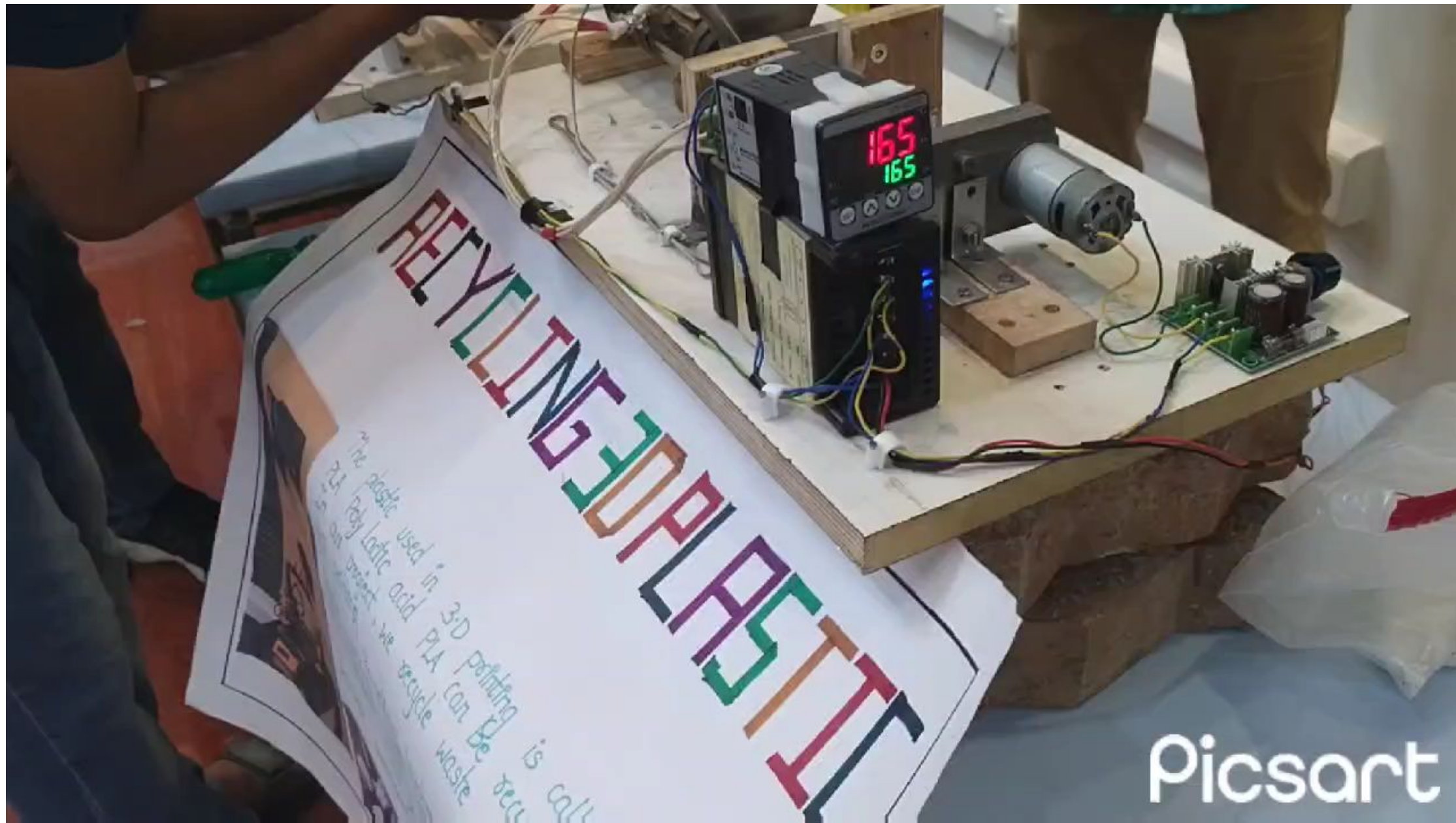
- We tried to figure them out.
- As mentioned above some of the hardware problems were- small size coupling, misalignment in the shaft, diameter of the filament was inconsistent, etc.
- Machining of coupling needed to be carried out using lathe machine with the support of personnel from Central Work Shop after getting permission letter signed from concerned authorities.
- The inner diameter of the coupling was increased from 12 mm to 12.55 mm.

OUR WORK AS A GROUP

- The motor base was in 3D plastic model, so it was not suitable for this project since it was of lower thickness and could not be fixed.
- So we went to Engineering practice laboratory took permission from concerned authorities and made a wooden model base frame suitable to support our shaft. It was of length 120 mm, thickness 14.9 mm.
- The existing L-shaped steel support and U-shaped wooden support were removed.
- Suitable sized holes were drilled using drilling machine obtained from DEE laboratory on U-shaped supports.
- Then the system were positioned properly and the filament extruder was assembled.

OUR WORK AS A GROUP

- The shaft was fit into the coupling, drive and driven shaft were aligned properly and bolted into the drilled holes.
- Understanding the terminations were difficult so we spent a bit of time on how to handle and operate them. Then we wired the plug top and made it suitable to our requirement.
- Then we connected 230 V, AC power supply, switched on the heater and increased the temperature till 160 °C slowly.
- Once the temperature reached around 150 to 160°C, we switched on the motor (connected to the coupler) due to which the plastic which was jammed inside the feed funnel melted and got extruded.
- We then shred PLA plastic supports in the shredding machine in the DEE laboratory safely after obtaining clearance from HOD, DEE. So that this can be used in future to extrude wire.



Picsart

Conclusion

- We all are really enjoying working on this project and developing key teamwork skills.
- We aptly **divided the work** between members **as per their skill sets**, likings, and abilities, (like few were good at understanding the wiring and setup for adjusting temperature, few have exceptional negotiation skills, few knew Tamil language, few were skilled at operating drill machines, etc.).
- So every person contributed a lot to the progress of this project, that too very **regularly** and **patiently**.

Conclusion

- We have developed the project from it's initial state to the present state where all its components are at correct positions and are operational.
- But a few things are still remaining to be done, which we plan to finish soon. Like, working on the consistency and quality of the filament by trying to run the machine at various temperatures and motor rpms. Also, we need to attaching a motor to the spool to role the newly made filament wire.



THANK YOU